

ARRAYS

Q9)

Question:

DS	Session	Arrays	Question Information	Level 1	Challenge 29
Problem Description: Tina has given a boolean matrix <code>mat[P][Q]</code> of size <code>P X Q</code> to Laaysa, She wants to modify it such that if a matrix cell <code>mat[m][n]</code> is 1 (or true) then make all the cells of <code>m_th</code> row and <code>nth</code> column as 1. Can you help Laaysa? Constraints: <code>1 <= p, q <= 1000</code> Input Format: First line of the input is the how many rows and columns in Laaysa matrix. After that, each line will represent each row and Laaysa need to enter numbers. Output Format: Print the resultant matrix output in a separate lines.					

PROGRAM:

```
1  #include <stdio.h>
2
3  int main() {
4      int p, q;
5      static int mat[1000][1000];
6      int row[1000] = {0};
7      int col[1000] = {0};
8
9      scanf("%d%d", &p, &q);
10
11
12     for (int i = 0; i < p; i++) {
13         for (int j = 0; j < q; j++) {
14             scanf("%d", &mat[i][j]);
15
16             if (mat[i][j] == 1) {
17                 row[i] = 1;
18                 col[j] = 1;
19             }
20         }
21     }
22     printf("\n");
23
24     for (int i = 0; i < p; i++) {
25         for (int j = 0; j < q; j++) {
26             if (row[i] == 1 || col[j] == 1) {
27                 mat[i][j] = 1;
28             }
29         }
30     }
31
32     for (int i = 0; i < p; i++) {
33         for (int j = 0; j < q; j++) {
34             printf("%d", mat[i][j]);
35
36             if (j < q - 1)
37                 printf(" ");
38         }
39         printf("\n");
40     }
41
42     return 0;
43 }
```

OUTPUT:

TEST CASE 1:

```
2 2
1 0
0 1
```

```
1 1
1 1
```

TEST CASE 2:

```
3 3
0 0 0
0 1 0
0 0 1
```

```
0 1 1
1 1 1
1 1 1
```